



Guidelines

Low Dose Heparin Infusion for Post-Revascularisation Procedures Guideline

Reference #: HRM031

Scope

Site	Service/Department/Unit	Disciplines
SCGH	G62 Vascular Surgery	Medical, Nursing, Pharmacy

HIGH RISK MEDICATION

1.0 Introduction

Complicated or extensive endovascular intervention is associated with a risk of early post-intervention thrombosis. This is particularly the case with extensive superficial femoral artery stenting and tibial interventions. Post-operative subtherapeutic low-dose heparin is sometimes administered in these cases to prevent early thrombosis which can have significant adverse outcomes¹.

2.0 Availability²

- All heparin concentrations: 1000 units/1 mL, 5000 units/0.2 mL, 5000 units/1 mL, 5000 units/5 mL, 25,000 units/5 mL
- Available in generic brands.

3.0 Contraindications^{2, 3}

- Known hypersensitivity to heparin or pork products.
- History of heparin induced thrombocytopenia and heparin induced thrombocytopenia and thrombosis.
- Patients with a platelet count less than $50 \times 10^9/L$ or patients for whom suitable blood coagulation tests cannot be performed at appropriate intervals.
- Current active bleeding.

4.0 Precautions^{2, 3}

- Patients at high risk of bleeding.
- Thrombocytopenia.
- Severe hepatic impairment or disease.
- Uncontrolled hypertension.

5.0 Indication

For post revascularisation procedures in patients that are deemed high risk of early thrombosis under the direction of Vascular Surgeon or Interventional Radiologist.

6.0 Dosage & Administration

NO BOLUS dose of heparin is to be given.

DO NOT use the WA Anticoagulation Medication Chart Nomogram

1. Dilute heparin 25,000 units in 500 mL sodium chloride 0.9% (50 units/mL)
2. Mix bag thoroughly (invert the bag at least 6 times to prevent pooling of heparin)^{2,4}
3. Administer as an intravenous infusion at a rate of 500 units/hour (10mL/hr)

7.0 Monitoring^{2, 3}

DO NOT monitor APTT unless requested by the Vascular Surgeon or the Interventional Radiologist or patient develops active bleeding.

- Platelet level should be obtained at baseline and every 24 hours whilst heparin infusion is running. A mild drop in platelet count may occur in within 1 to 4 days of heparin infusion, whilst a delayed, more severe form of thrombocytopenia may occur 5 to 14 days after starting heparin. Refer to the [Management of Heparin Induced Thrombocytopenia \(HIT\) guideline](#) for further information.
- Monitor for signs and symptoms of bleeding.
- Immediate hypersensitivity reactions i.e., chills, fever and urticaria
- Skin necrosis at injection sites may be due to heparin-induced platelet aggregation and thrombosis. Stop heparin infusion immediately.

8.0 Nursing Management

- Infusion should be made up as per medical order.
- Infusion to be commenced (time/duration) as documented by Vascular Surgeon or Radiological Interventionalist.
- Antiplatelet agents (aspirin and clopidogrel) may be given concurrently as prescribed.
- Perform post angiogram neurovascular observations as per standard angiographic protocols.
- Contact Vascular Registrar via switch if concerns at any stage.

9.0 Practical Points

9.1 Prescribing

A Low Dose Heparin infusion sticker should be affixed to the WA Anticoagulation Medication Chart under the heading 'Maintenance Infusion Rate Changes and Bolus Doses' to indicate that fixed dose of heparin is being used (see Appendix 1. Low Dose Heparin Infusion sticker).

- Any prescription of heparin should be charted on the WA Anticoagulation Medication Chart MR401/805.2

9.2 Drug Interactions^{2,3}

- Anticoagulants and/or antiplatelets:** concomitant use can increase bleeding risk. If antiplatelets are indicated, monitor for any signs and symptoms of bleeding.
- Intravenous glyceryl trinitrate:** may decrease partial thromboplastin time with a subsequent rebound effect upon discontinuation of the glyceryl trinitrate.
- Medications that increase serum potassium: concomitant use with drugs that increase serum potassium (such as **ACE inhibitors, spironolactone, amiloride**) may lead to significant increases in serum potassium.
- Digoxin** may partially counteract the anticoagulant action of heparin. If concomitant use is required, check with the vascular medical team if an APTT is required.

9.3 Adverse Effects^{2,3}

- Bleeding
- Bruising and pain at injection site
- Hyperkalaemia
- Mild reversible thrombocytopenia
- Severe thrombocytopenia
- Skin necrosis (usually at injection site)
- Allergic reactions including urticaria and/or anaphylaxis

9.4 Compatibility⁴

- Compatible fluids:** Glucose 5%, sodium chloride 0.9%
- Y site compatibility:** Aciclovir, amifostine, aminophylline, ampicillin, anidulafungin, atropine, azathioprine, aztreonam, bivalirudin, caffeine citrate, calcium chloride, calcium gluconate, cefazolin, cefotaxime, cefoxitin, ceftaroline, ceftazidime, ceftolozane-tazobactam, ceftriaxone, cefuroxime, cyclosporin, clindamycin, defibrotide, dexamethasone, digoxin, edrophonium, ephedrine sulfate (adrenaline), eptifibatide, ertapenem, esmolol, ethanol, fentanyl, fluconazole, foscarnet, furosemide, ganciclovir, granisetron, hydralazine, hydrocortisone sodium succinate, hyoscine hydrobromide, insulin (Novorapid), levetiracetam, lidocaine, linezolid, magnesium sulfate, metoclopramide, meropenem, metoprolol, metronidazole, micafungin, midazolam, morphine sulfate, neostigmine, oxytocin, palonosetron, paracetamol, phenylephrine, piperacillin-tazobactam, remifentanyl, sodium bicarbonate, tigecycline, tirofiban, zidovudine
- Incompatible with the following drugs:** Alteplase, amiodarone, benzylpenicillin, capreomycin, caspofungin, ciprofloxacin, cisatracurium, desferrioxamine, dobutamine, droperidol, ergometrine, erythromycin, filgrastim, gentamicin, haloperidol lactate, hyaluronidase, ketamine, labetalol, levomepromazine, methadone, mycophenolate mofetil, palifermin, pentamidine, pethidine, promethazine, protamine, quinine, sildenafil, tobramycin, tramadol

9.5 Storage^{2,4}

- **Ampoules:** store below 25°C (room temperature)

9.6 Pregnancy and Lactation^{2,3,5}

- **Pregnancy:** Category C. Heparin is considered safe to use during pregnancy, however the use of anticoagulants during pregnancy may be associated with increased risk of placental haemorrhage and subsequent pre-term birth or foetal loss. Consultation with a haematologist for further advice is recommended.
- **Lactation:** Safe to use

Related Documents

Legislation

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)

Standards, procedures, guidelines:

- [Management of Heparin Induced Thrombocytopenia \(HIT\)](#)
- [SCGH Nursing Practice Guideline No.51 Medication Management](#)
- [SCGH Nursing Practice Guideline No.4 Intravenous Therapy](#)
- NSQHS Standards: 1,4

References

1. Jayarajan SN, Holzem KM, Desai K, Sanchez LA, Zayed MA. Postoperative Low-Dose Heparin Infusion Does Not Change Complication Rates after Limb Revascularization. *Annals of Vascular Surgery*. 2019;56:280-6.
2. MIMS Online [Internet]. St Leonards: MIMS Australia; 2023 [modified 2021 April 1; cited 2023 July 28]. Available from: <https://www.mimsonline-com-au.qelibresources.health.wa.gov.au/Search/Search.aspx>
3. Rossi S. Australian Medicines Handbook [Internet]. Adelaide: Australian Medicines Handbook; 2023 [modified 2023 July 1; cited 2023 July 28]. Available from <https://amhonline-amh-net-au.qelibresources.health.wa.gov.au/>.
4. Burrridge N, Symons K. Australian Injectable Drugs Handbook [Internet]. Collingwood: The Society of Hospital Pharmacists of Australia; 2023 June [cited 2023 July 28]. Available from: https://aidh-hcn-com-au.qelibresources.health.wa.gov.au/browse/h/heparin_sodium
5. Micromedex [Internet]. USA: Merative L.P.: Micromedex; 2023 [modified 2023 June; cited 2023 July 28]. Available from: https://www.micromedexsolutions-com.qelibresources.health.wa.gov.au/micromedex2/librarian/CS/2A7C4E/ND_PR/evidencexpert/ND_P/evidencexpert/DUPLICATIONSHIELDSYNC/10B5E3/ND_PG/evidencexpert/ND_B/evidencexpert/ND_AppProduct/evidencexpert/ND_T/evidencexpert/PFActionId/pf.HomePage?navitem=topHome&isToolPage=true

Appendix 1

LOW DOSE HEPARIN INFUSION

Department of Vascular Surgery

Indication: Prevent graft/stent occlusion post operatively

INFUSION RATE

Fixed Rate: 10mL/hour (500 units/hour)

***** NO BOLUS TO BE GIVEN*****

*****NO APTT TO BE MEASURED: NON PROTOCOL*****

*****DO NOT USE NOMOGRAM: NON PROTOCOL*****

Figure 1. Low dose heparin infusion sticker to affix to WA Anticoagulation Medication Chart under the heading 'Maintenance Infusion Rate Changes and Bolus Doses' to indicate that fixed dose of heparin is being used.

Authorisation & Version Control

Policy Sponsor	Director of Clinical Services				
Policy Contact	Drugs and Therapeutics Committee Executive Officer (Senior Pharmacist)				
Date First Issued:	11/05/2020	Last Reviewed:	14/08/2023	Review Date:	14/08/2026
Version No.	2				
Approved by:	SCGOPHCG Drugs and Therapeutics Committee			Date:	22/08/2023
Acknowledgments:	SCGH Pharmacy, Head of Department Vascular Surgery, Nurse Practitioner Vascular Surgery, Head of Department Radiology, Co-ordinator of Nursing Perioperative Services Theatre				
Standards Applicable	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.